Chimie organique 1

Exercices et solutions :

I. Exercices :

Exercice 1 :

Dessinez la formule développée des molécules suivantes :

a) $(CH_3)_2CHCH_2OH$

b) $Cl_2C=CCl_2$

c) $CH_3CCl_2CH_3$

d) $(CH_3)_2C(CH_2CH_3)_2$

Exercice 2 :

Dessinez la formule topologique (zig-zag) des molécules ci-dessous :

a) $CH_3(C=CH_2)CH_2CH_3$

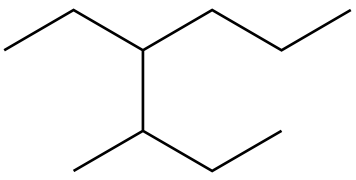
b) $CH_3C(CH_2CH_3)CH_2CH_3$

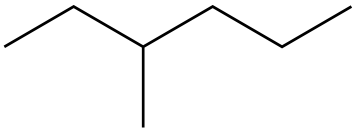
c) $CH_3CH_2CH=CHCH_2CH(CH_3)_2$

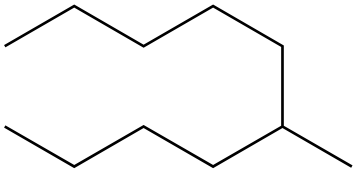
d) $(CH_3)_2CHCH_2CH(CH_3)_2$

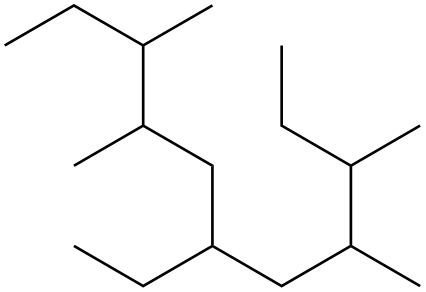
Exercice 3 :

Déterminez la chaine principale et les ramifications des molécules ci-dessous :

a) 

b) 

c) 

d) 

Exercice 4 :

Ecrivez une formule simplifiée ou stylisée contenant :

a) Un homocycle ou composé carboxylique de 5 C.

b) Un hétérocycle ou composé hétérocyclique avec deux ramifications.

c) Une chaîne principale de 7 C avec 2 ramifications de 2 C chacune.

d) Une chaîne principale de 8 C avec une ramification méthyle et une ramification isopropyle.

Exercice 5 :

Dessinez en formule topologique (zig-zag) les structures associées aux noms suivant selon les règles IUPAC.

a) 4-isopropyl-3-méthylheptane.

b) 3,7-diéthyl-5-isopropyldécane.

c) 2,3,5-triméthylheptane.

d) 4-éthyl-2-méthylhexane.

Exercice 6 :

Corrigez, s'il y a lieu, le nom des substances suivantes :

a) 4-éthyl-2-méthylpentane.

b) 5-éthyl-3-méthylheptane.

c) 3-isobutyl-5-isopropylheptane.

d) 4-(sec-butyl)-6-(tert-butyl)-3,5-diéthyl-7-isopropyl-2-méthylnonane.

Exercice 7 :

Donnez la formule semi-développée ou le nom des produits suivants :

a) hexa-1,3-diène

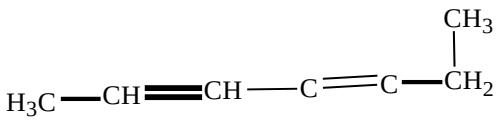
b) hept-2-yne-4-ène

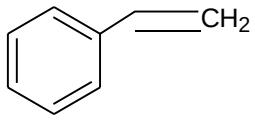
c) pentène

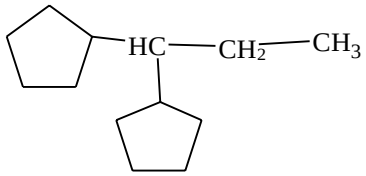
d) octa-1,3,5-trién-7-yne

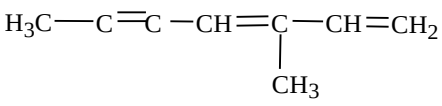
Exercice 8 :

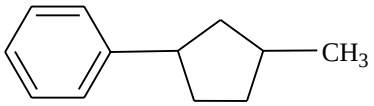
Nommer les molécules ci-dessous **:**

a) 

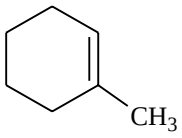
b) 

c) 

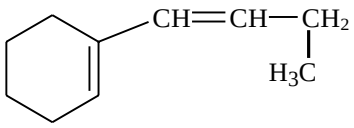
d) 

e) 

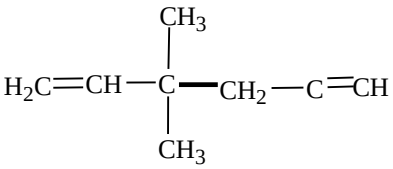
f)


CH3

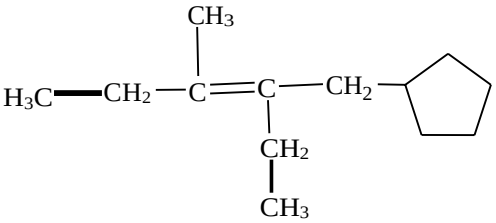
g)


CH
CH
CH2
H3C

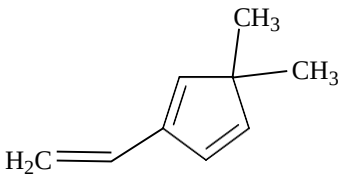
h)


CH3
H2C
CH
C
CH2
C
CH
CH3

i)


CH3
H3C
CH2
C
C
CH2
CH2
CH3

j)


CH3
CH3
H2C

Exercice 9 :

Dessiner les molécules ci-dessous :

a) 3-cyclohex-1-énylbutène
b) 1-buta-2,3-diénylpent-1-yne
c) 2-prop-2-ényl-1-méthylbenzène
d) 2,3-diméthylnona-5,6,8-triène
e) 1,3-dicyclohex-2-énylprop-1-ène
f) 1-phénylbut-1-yne
g) 2-cyclopentylbut-3-ène

Exercice 10 :

Dessiner les molécules ci-dessous :

a) 1-penta-1,4-diénylbut-2-yne
b) 1,3-dicyclopent-2-énylbut-2-ène
c) 2-cyclopent-2-énylbutène
d) 2,3,3-triméthylocta-4,5,7-triène
e) 2-pentylbut-3-ène
f) 2,2-dicyclopent-2-ényl-4-méthylhept-6-ène
g) 2-phénylhex-5-yne
h) 2,5-diméthylhexa-1,4-diène
j) 2-éthyl-3-hex-4-énylbenzène

Exercice 11 :

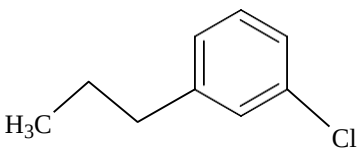
Dessinez en formule topologique (zig-zag), les molécules suivantes :

a) 3,3,4-triméthylnona-1,6-diyne.
b) 3-tert-butyl-4-chlorohex-1-én-5-yne.
c) 3-sec-butylhept-5-én-1-yne.

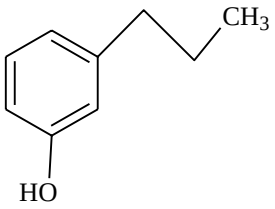
Exercice 12 :

Indiquez le nom IUPAC des molécules suivantes :

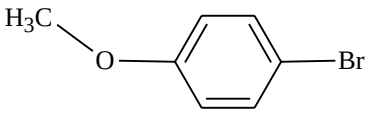
a)



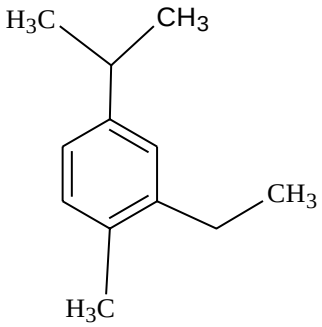
b)



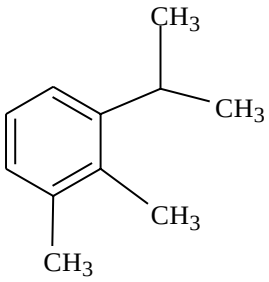
c)



d)



e)



Exercice 13 :

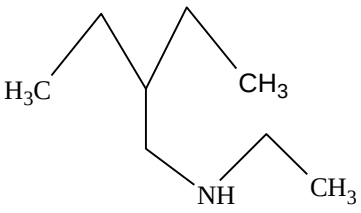
Dessinez, en formule topologique (zig-zag), les structures correspondantes aux noms suivants :

a) 2-méthoxyoctane.
b) 3-éthoxy-4-éthylhexane.
c) 4-chloro-6-hexylhex-1-ène.
d) 5-cyclopentyloxycyclopenta-1,3-diène

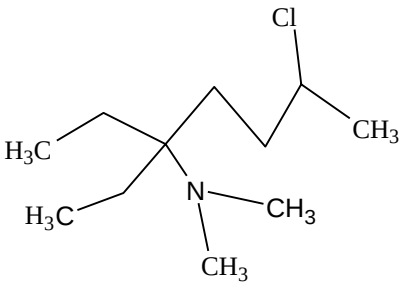
Exercice 14 :

Nommez les molécules suivantes selon les règles de l'IUPAC :

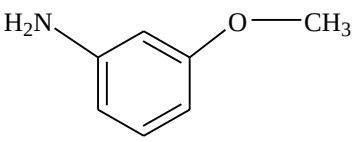
a)



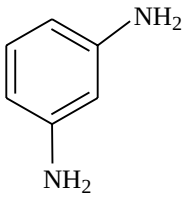
b)

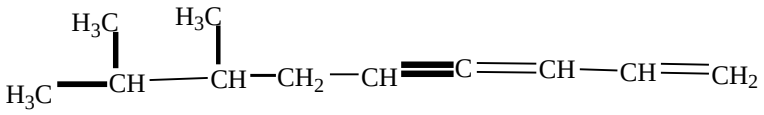


d)



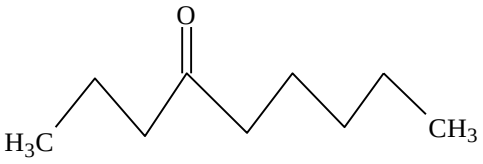
c)



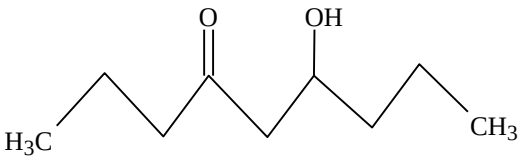
e) 

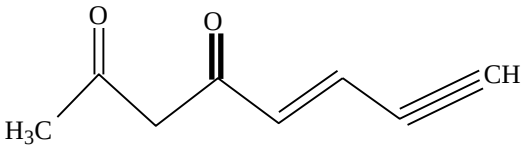
Exercice 15 :

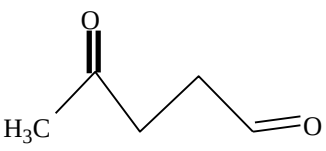
Nommez les molécules suivantes selon les règles de l'IUPAC :

a) 

b) 

c) 

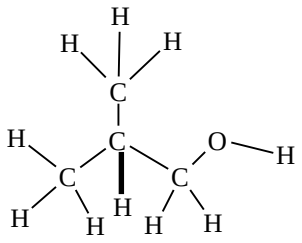
d) 

e) 

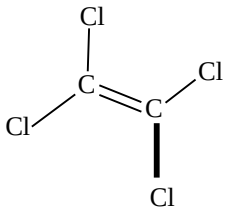
II. Solutions :

Exercice 1 :

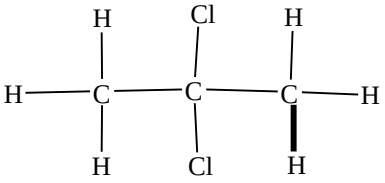
a)



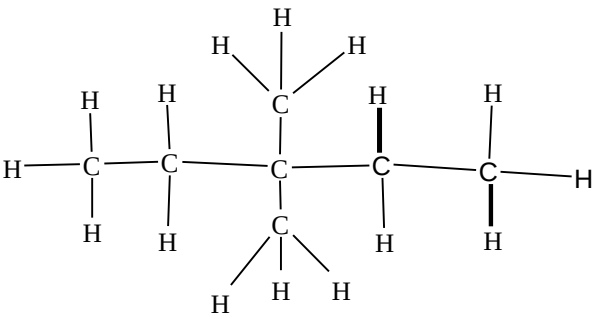
b)



c)

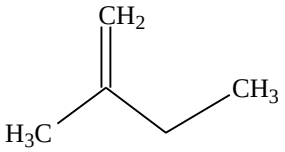


d)

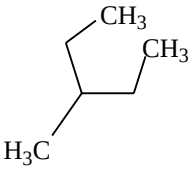


Exercice 2 :

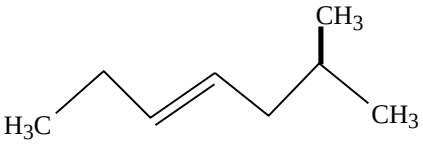
a)



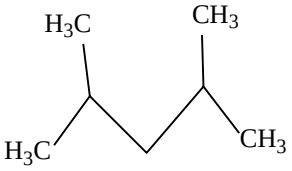
b)



c)



d)

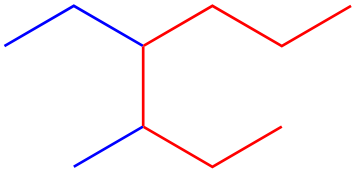


Exercice 3 :

Les chaînes principales sont dessinées en rouge, les substituants en bleu :

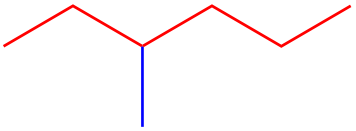
Chaîne principale : 7 carbones heptane.2 substituants : méthyle et éthyle.

a)



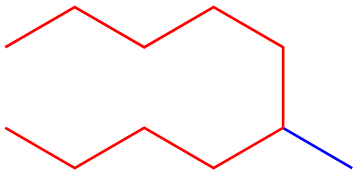
Chaîne principale : 6 carbones
hexane.1 substituant : méthyle.

b)



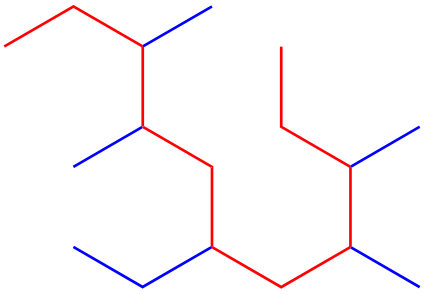
Chaîne principale : 10 carbones décane.1
substituant : méthyle.

c)



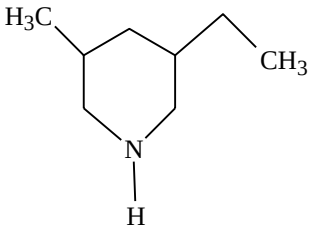
Chaîne principale : 11 carbones undécane.5
substituants : 4 méthyle et 1 éthyle.

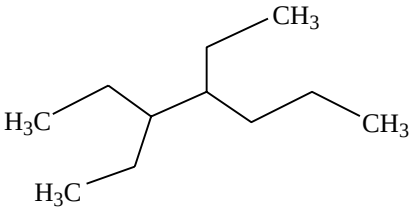
d)

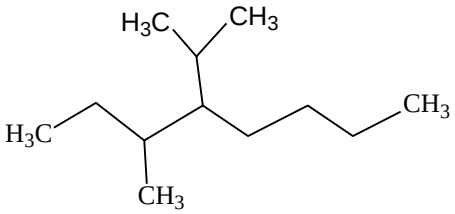


Exercice 4 :

a) 

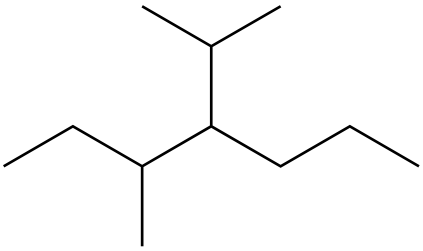
b) 

c) 

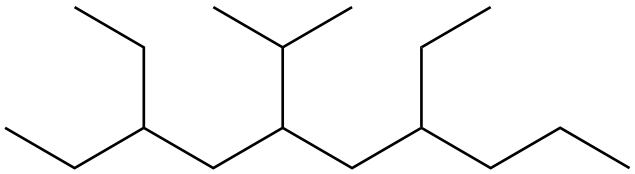
d) 

Exercice 5 :

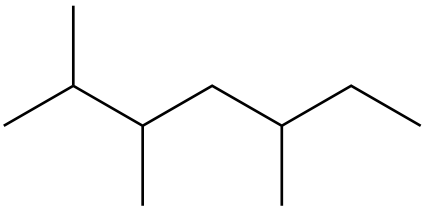
a)



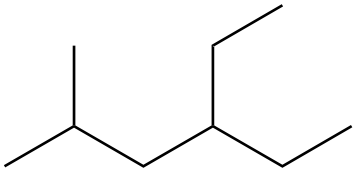
b)



c)

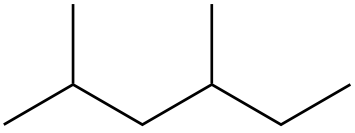


d)



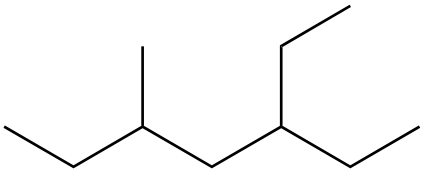
Exercice 6 :

a)



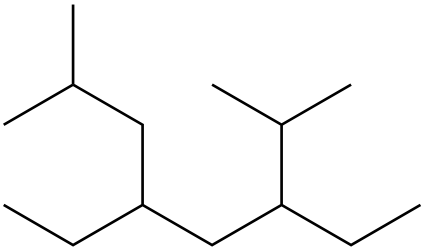
Nom correct : 2,4-diméthylhexane

b)



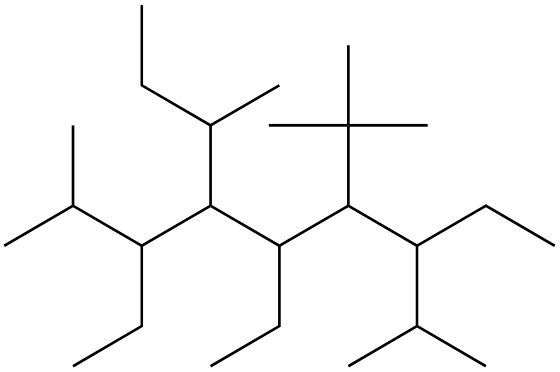
Nom correct : 3-éthyl-5-méthylheptane

c)



Nom correct : 3,5-diéthyl-2,7-diméthyloctane.

d)



Nom correct: 4-(*sec*-butyl)-6-(*tert*-butyl)-3,5,7-triethyl-2,8-dimethylnonane.

Exercice 7 :

a) $H_2C=CH-CH=CH-CH_2-CH_3$

b) $H_3C-C\equiv C-CH=CH-CH_2-CH_3$

c) $H_2C=CH-CH_2-CH_2-CH_3$

d) $H_2C=CH-CH=CH-CH=CH-CH=CH-C\equiv CH$

Exercice 8 :

a) hept-2-én-4-yne

b) phényléthène

c) 1,1-dicyclopentylpropane

d) 3-méthylhepta-1,3-diène-5-yne

e) 1-méthyl-3-phénylcyclopentane

f) 1-méthylcyclohex-1-ène (ou méthylcyclohexène)

g) 1-but-1-énylcyclohex-1-ène (ou 1-cyclohex-1-énylbut-1-ène)

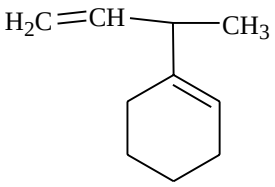
h) 3,3-diméthylpent-1-én-4-yne

i) (2-éthyl-3-méthylpent-2-én-1-yl)cyclopentane

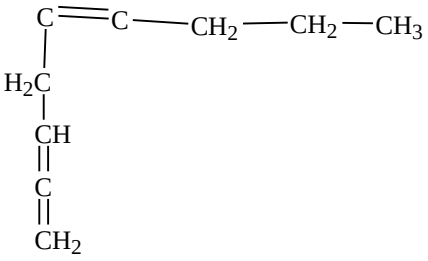
j) 5,5-dimethyl-2-vinylcyclopenta-1,3-diène

Exercice 9 :

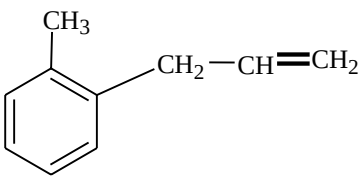
a)



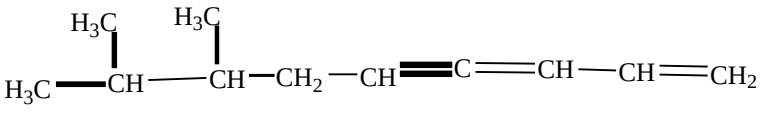
b)



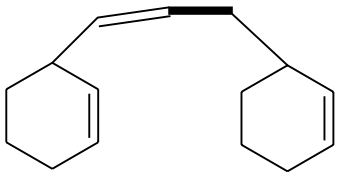
c)



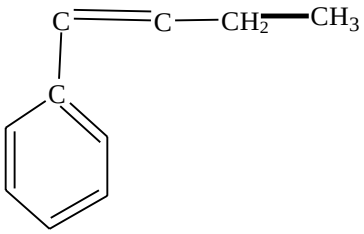
d)



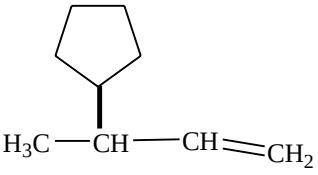
e)



f)

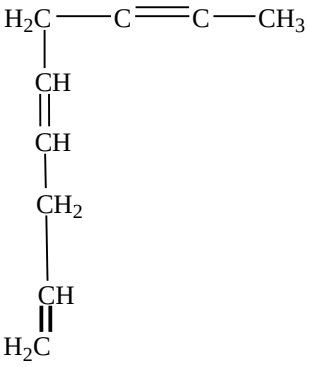


g)

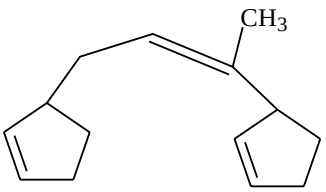


Exercice 10 :

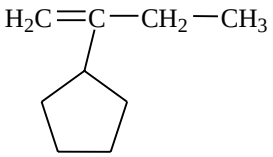
a)



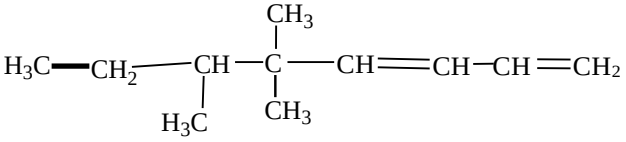
b)



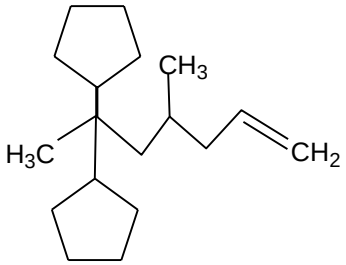
c)



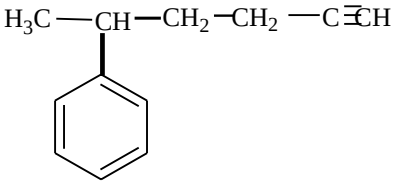
d)



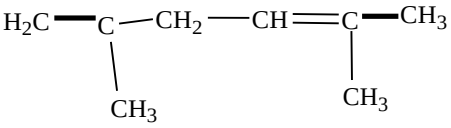
f)



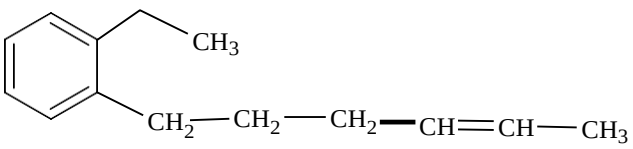
g)



h)

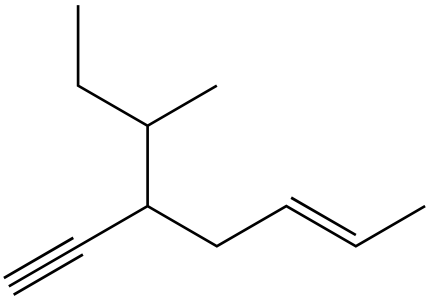


J)

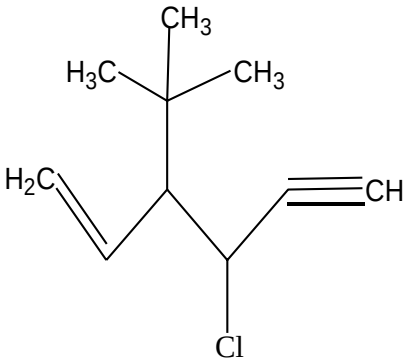


Exercice 11 :

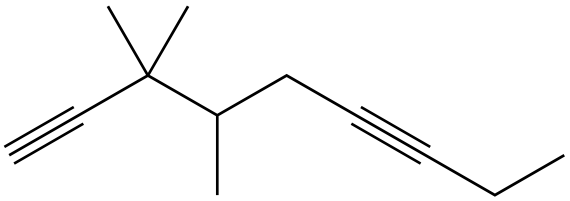
a)



b)



c)

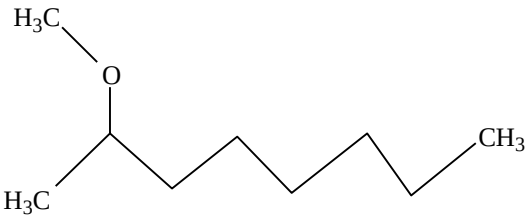


Exercice 12 :

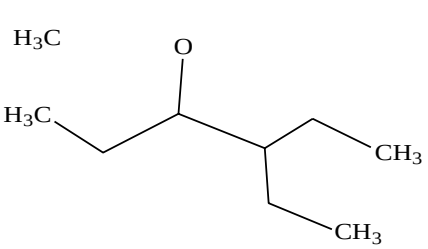
a) 1-chloro-3-propylbenzene.
b) 3-propylphénol.
c) 1-bromo-4-méthoxybenzène.
d) 2-éthyl-4-isopropyl-1-méthylbenzène.
e) 1-isopropyl-2,3-diméthylbenzène.

Exercice 13 :

a)



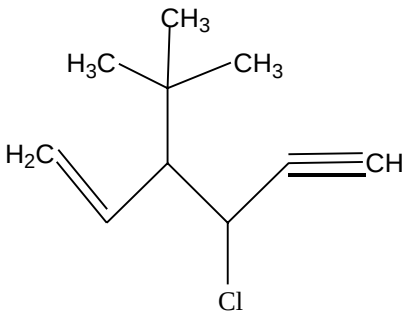
b)



c)



d)



Exercice 14 :

e) N,2-diéthylbutan-1-amine.
f) 6-chloro-3-éthyl-N,N-diméthylheptan-3-amine.
g) Cyclohexane-1,3-diamine.
h) 3-méthoxyaniline.
i) 3-amino-7,7-dimethyloct-1-yn-4-ol.

Exercice 15 :

a) Nonan-4-one.
b) Oct-5-én-7-yn-2-one
c) 6-hydroxynonan-4-one.
d) Oct-5-en-7-yne-2,4-dione.
e) 4-oxopentanal.